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RESEARCH

VEMORA: Vaginal Estrogen versus non-hormonal MOisturizer in women Receiving Aromatase inhibitors: a randomized, controlled trial

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Abstract

Purpose Genitourinary Syndrome of Menopause (GSM) remains a significant unmet need for postmenopausal breast cancer survivors receiving aromatase inhibitor (AI) therapy.

Methods In this randomized, controlled, open-label clinical trial, ER+ breast cancer survivors receiving adjuvant AI therapy with symptomatic GSM were randomized to receive 24 weeks of a non-hormonal vaginal moisturizer (Replens®) or vaginal estrogen therapy (either Vagifem® vaginal tablets or Estring® vaginal ring). The primary endpoint was change in vaginal dryness. Secondary endpoints included dyspareunia, sexual functioning, vaginal pH, and serum estradiol values.

Results Twenty-three patients were enrolled (vaginal estrogen [VE], $n=11$; non-hormonal [NH], $n=12$). Vaginal dryness and dyspareunia scores improved significantly in the VE arm compared with the NH arm ($p=0.049$, $p=0.048$, respectively). Vaginal pH decreased significantly with vaginal estrogen compared to non-hormonal therapy ($p=0.0286$). Other domains of sexual function (libido, ability to achieve orgasm, sexual satisfaction, and relationship with partner) were not significantly different between the 2 groups. Serum estradiol levels remained low through the first 12 weeks of therapy. Two participants demonstrated estradiol elevations above threshold at 24 weeks (21.8 pg/mL and 16 pg/mL); both reported non-adherence to AI therapy preceding measurement.

Conclusion In this randomized study, vaginal estrogen provided greater improvement in vaginal dryness and dyspareunia than non-hormonal therapy in breast cancer survivors. These findings provide prospective evidence supporting symptomatic benefit with minimal systemic estrogen exposure in most patients, although larger studies with longer follow up are needed to define long-term safety and efficacy.

Keywords Hormone receptor positive breast cancer · Genitourinary syndrome of menopause · Aromatase Inhibitor-Induced side effects · Vaginal estradiol treatment · Breast cancer survivorship

Introduction

Breast cancer is the most common malignant disease in women in the Western world. Approximately 310,720 women in the US are diagnosed with invasive breast cancer every year, and more than 42,000 women die of this disease. The vast majority of breast cancer cases (84%) occur in women aged 50 years and older, with most of these women being post or peri-menopausal [1]. Within this group of women, approximately 80% of breast cancer cases are hormone receptor positive (estrogen receptor, ER; progesterone receptor, PR) [2]. As dictated by the standard of care, most post-menopausal women with hormone receptor positive breast cancer will be treated with adjuvant aromatase inhibitor (AI) therapy [3–6]. Unfortunately, because AI's decrease estradiol levels to nearly zero, more than 60% of post-menopausal breast cancer survivors experience genitourinary syndrome of menopause (GSM), previously termed atrophic vaginitis or vulvovaginal atrophy [7]. This results in decreased vaginal moisture and elasticity, which typically leads to dyspareunia. Moreover, low estrogen levels

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promote higher vaginal pH (ranging from 5.5 to 6.8) which causes loss of lactobacilli and overgrowth of other bacteria, thus increasing risk of urinary tract infections [8, 9].

Indeed, for breast cancer survivors on AI therapy, vaginal dryness and sexual dysfunction are significant problems. Data from the ATAC trial shows that 18.5% of women on anastrazole have vaginal dryness, 34% have diminished libido, and 17.3% have dyspareunia [10]. Currently, the most common treatment for GSM in ER+ breast cancer survivors has been non-hormonal (NH) polycarbophil-based vaginal moisturizers, such as Replens®, which has been studied in several clinical trials [11–13]. However, when compared to placebo cream, NH cream has shown no benefit in vaginal dryness or dyspareunia [12]. Furthermore, when compared to vaginal estrogens, NH cream was found to be inferior to estriol vaginal cream in improving sexual function, with symptoms returning to pre-treatment values by week 12 when using NH cream [11]. Nonetheless, for lack of a better alternative, oncologists commonly suggest topical polycarbophil-based vaginal moisturizers for GSM among their breast cancer survivors.

In the non-breast cancer population, vaginal estrogens have been proven to be effective for GSM, with minimal to no elevation in serum estradiol level [14–16]. However, these medications have not been extensively studied in postmenopausal ER+ breast cancer patients on aromatase inhibitor therapy, due to fear of inadvertently elevating serum estradiol levels. A systematic review and meta-analysis in 118,659 breast cancer survivors where 6,358 women used vaginal estrogen showed no significant increase in risk of breast cancer recurrence (RR = 0.87, 95% CI: 0.67–1.11). In fact, users of vaginal estrogen had a significant reduction in all-cause mortality (RR = 0.80; CI 95%: 0.75–0.86) [17]. Most literature so far contains only observational studies. Hence, there is an unmet need to conduct prospective randomized studies studying the effects of vaginal estrogen in breast cancer survivors, especially those with hormone positive disease.

In this randomized, controlled trial, we examine the effects of vaginal estrogens versus NH cream for women with genitourinary syndrome of menopause on AI therapy. We report patients' subjective symptoms of vaginal dryness, sexual health, vaginal pH, and ultra-sensitive serum estradiol levels while on treatment.

Methods

Patients on adjuvant aromatase inhibitor therapy for stage I-III hormone receptor positive breast cancer were eligible for the trial if they were experiencing GSM. All patients had to be post-menopausal, defined as any of the following: age > 55 years; history of bilateral oophorectomy; amenorrhea for 1 year with intact uterus and ovaries; serum estradiol and FSH concentrations in the post-menopausal range along with either amenorrhea for 6 months or previous hysterectomy; or medical ovarian suppression with a GnRH agonist. Patients were excluded if they used any exogenous estrogen within four weeks of initial screening or if they had an active vaginal infection.

Patients were randomized 1:1 to receive either non-hormonal cream (Replens®) or vaginal estrogen (VE) for 24 weeks. Patients in the NH arm were asked to apply the vaginal moisturizer three times per week. The patients randomized to the VE arm could choose between Estring® (vaginal ring, 2 mg estradiol, replaced every 3 months) or Vagifem® (10 mcg tablets, placed vaginally daily for 2 weeks, then twice weekly), based on their own personal preferences. Allowing patient choice was intended to reflect real-world clinical practice and improve external validity.

At baseline and end-of-study, all patients had a gynecologic exam with vaginal pH measurement, as well as serum estradiol measurements via an ultra-sensitive, liquid chromatography/tandem mass spectrometry (LC-MS/MS) assay, with detection limits down to < 2 pg/mL. For those patients who were randomized to vaginal estrogen, they also had estradiol levels checked at weeks 2, 4, 12, and 24. Finally, all patients were asked to fill out a one-page vaginal dryness and sexual health questionnaire (see supplementary appendix A) at baseline, week 12, and at the end of the study.

For patients who had serum estradiol levels > 27 pg/mL at 2 weeks or > 12 pg/mL for the remainder of the study, the vaginal treatment would be stopped and not resumed. A brief surge in estradiol level at 2 weeks was allowed as previous data shows that this surge, if it occurs, is transient and self-resolving [18]. The limit of 12 pg/mL was selected as a conservative threshold exceeding expected estradiol levels during effective aromatase inhibition, while remaining below typical postmenopausal concentrations [19].

This study was conducted in accordance with principles of Good Clinical Practice. The study protocol was approved by the institutional review boards at Baylor College of Medicine and Houston Methodist Hospital (IRB number 00017710). Written informed consent was obtained from all patients before participation in the study. The trial was registered at Clinicaltrials.gov (NCT 01984138) on October 8. 2013. The trial was monitored by the Institutional Data and Safety Monitoring Committee at Houston Methodist Hospital to ensure participant safety and adherence to ethical and regulatory standards.

Statistics

The primary endpoint was improvement in vaginal dryness, as measured by the validated vaginal dryness questionnaire. Secondary endpoints included sexual functioning (questionnaire-based) and vaginal pH.

Sample size calculation is based on previous estimates of the effects of both vaginal estrogen (mean change=-11.6 score from baseline to 12 weeks, σ = 5.2) and vaginal moisturizer (mean change=-1.3, σ = 5.5) on vaginal atrophy [11]. However, the scores obtained in that study were derived from a different questionnaire. Thus, rather than using absolute scores, we constructed our statistical plan based on the effect size. The vaginal symptom scores in the Biglia study differed by 1.9 standard deviations at the 12 week endpoint. Because our study is a randomized controlled study rather than a retrospective study, and because both the vaginal estrogen concentration and serum estrogen concentration of Estring is lower than the vaginal cream and tablet used in the Biglia study, we will be slightly more conservative than the findings of the Biglia study. Group sample sizes (Vaginal Estrogen or Replens) of 9 and 9 achieve 84.761% power to reject the null hypothesis of zero effect size when the population effect size is 1.5 and the significance level (alpha) is 0.05 using a two-sided two-sample equal-variance t-test. Accounting for up to 20% drop out rate, we planned to accrue a total of 22 patients to the trial.

An intent-to-treat analysis on all patients who start vaginal estrogen or Replens was used for the primary efficacy analysis. A composite vaginal dryness score was calculated based on the average of the vaginal dryness and vaginal itching specified on the questionnaire. The primary endpoint is the difference in the scores between baseline and Week 24 between the two treatment groups. For those patients who are taken off the study prematurely for any reason, we used the difference between the first and last recorded vaginal dryness score. A two-sample Wilcoxon rank-sum test was used to examine the effects of the treatment on GSM (as defined by vaginal dryness and vaginal itching).

Table 1 Patient characteristics

	Overall (N=23)	Vaginal Estrogen (N=11)	Non-hormonal (N=12)
Median age:	56	56	57
Age Range	44–78	44–78	47–72
Race			
American Indian	1/23 (4%)	0	1/12 (8%)
Black/African American	2/23 (9%)	1/11 (9%)	1/12 (8%)
White	19/23 (83%)	10/11 (91%)	9/12 (75%)
Declined to state	1/23 (4%)	0	1/12 (8%)
Ethnicity			
Hispanic	3/23 (13%)	0	3/12 (25%)
Non-Hispanic	20/23 (87%)	11/11 (100%)	9/12 (75%)
Menopausal status			
Post-menopausal	17/23 (74%)	8/11 (72.7%)	9/12 (75%)
Pre-menopausal, on ovarian suppression	6/23 (26%)	3/11 (27.3%)	3/12 (25%)
Average Weight	72.5 Kg	69.9 Kg	75.0 Kg
Weight Range	50.3–107 Kg	59.6–107 Kg	50.3–101 Kg
Average BMI	27.6	26.1	28.9
BMI Range	19.1–40.7	19.1–38.8	20.3–40.7
Stage			
Stage I	7 (30%)	2 (18%)	5 (42%)
Stage II	13 (57%)	7 (64%)	6 (50%)
Stage III	3 (13%)	2 (18%)	1 (8%)

Results

Twenty-three patients were enrolled in the study, with 12 patients assigned to the non-hormonal moisturizer arm, and 11 patients assigned to the vaginal estrogen arm. Of the subjects who received vaginal estrogen, 8 chose the vaginal ring, and 3 chose the vaginal tablet. See Table 1 for additional details. All randomized participants initiated assigned therapy and were included in the intent-to-treat analysis.

Clinical outcomes

There was a significant improvement in the composite vaginal dryness score for those patients in the vaginal estrogen arm, as compared to those in the non-hormonal moisturizer arm. The questionnaire scale was from 0 to 4, with 0 representing no vaginal symptoms, and 4 representing very severe vaginal symptoms. The patients in the vaginal estrogen arm saw an average improvement of 1.6 points, compared to a 0.9 point improvement in the non-hormonal arm (p=0.049). There was also an improvement in patient-assessed dyspareunia, assessed on a scale from 0 to 10, with a score of 10 representing extremely painful intercourse. Women on the estrogen arm saw an improvement of 4.8 points, whereas those on the non-hormonal treatment arm improved by an average of 1.4 points (p=0.017). However, the improvement in other sexual functioning parameters (libido, ability to achieve orgasm, sexual satisfaction, and relationship with partner) was not significantly different between the two groups. See Table 2.

Similarly, there was significantly increased reduction in vaginal pH in the vaginal estrogen arm, indicating return to a healthier vaginal environment. The mean reduction in pH in the estrogen arm was 1.67, compared to a reduction of 0.36 in the non-hormonal arm (p = 0.0286). See Fig. 1.

Table 2 Vaginal/sexual health differences in NH and VE arms from baseline to end of study

	NH Moisturizer		Vaginal Estrogen		*p* value
	Difference	#Patients	Difference	#Patients	
Vaginal Dryness	-0.9	12	-1.6	11	*0.049
Dyspareunia	-1.4	7	-4.8	9	*0.017
Libido	-1.4	7	-0.4	10	0.182
Orgasm	-2.7	6	-1.8	9	0.338
Sexual Satisfaction	-2	6	-2	9	0.5
Partner Relationship	-0.5	8	-0.3	10	0.449

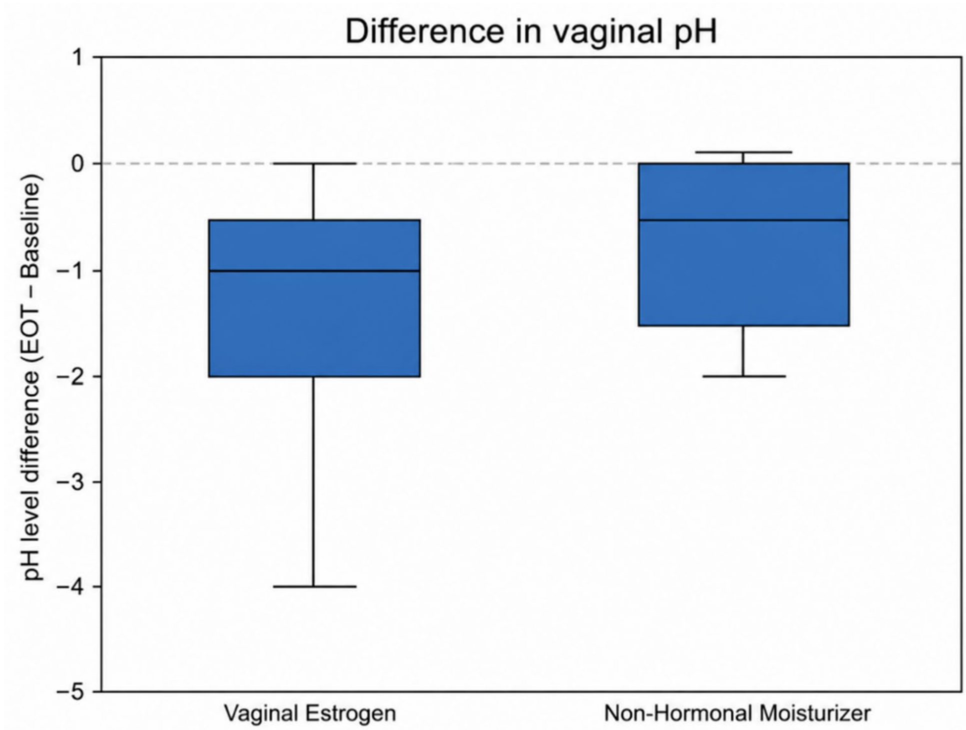

Fig. 1 Difference in vaginal pH between vaginal estrogen and non-hormonal moisturizer groups

Hormonal outcomes

Of the eleven patients on the vaginal estrogen treatment, only two patients had any estradiol values > 5 pg/mL in weeks 2 or 4, with one patient showing estradiol level of 8.6 pg/mL at 2 weeks, and the other patient showing a value of 6.9 pg/mL at 4 weeks. No participants demonstrated estradiol values exceeding the prespecified threshold during the initial treatment period. However, at week 24, two patients on the vaginal estrogen arm had estradiol levels greater than 10 pg/mL (21.8 and 16 pg/mL). See Fig. 2. Both participants were naturally post-menopausal, and they reported non-adherence to aromatase inhibitor therapy prior to measurement. One of these patients was using the estrogen vaginal ring, and the other was using the estrogen vaginal tablet.

Discussion

In this randomized controlled trial of breast cancer survivors receiving adjuvant aromatase inhibitor therapy, vaginal estrogen resulted in greater improvement in vaginal dryness, dyspareunia, and vaginal pH compared with a non-hormonal moisturizer. Though the more objective measures of vaginal pH and dyspareunia were significantly improved in the vaginal estrogen arm, the more subjective measures of sexual and relationship health (libido, orgasm, sexual satisfaction, and partner relationship) were not significantly improved. While the sample size was small, and only a subset of this small group was sexually active, this pattern does highlight the fact that sexual health is a very complicated, multi-factorial issue, which may not necessarily improve with simply treating vaginal dryness. Other factors, such as overall health, self-image, body changes, stress level, and relationship dynamics play an important role in these measures.

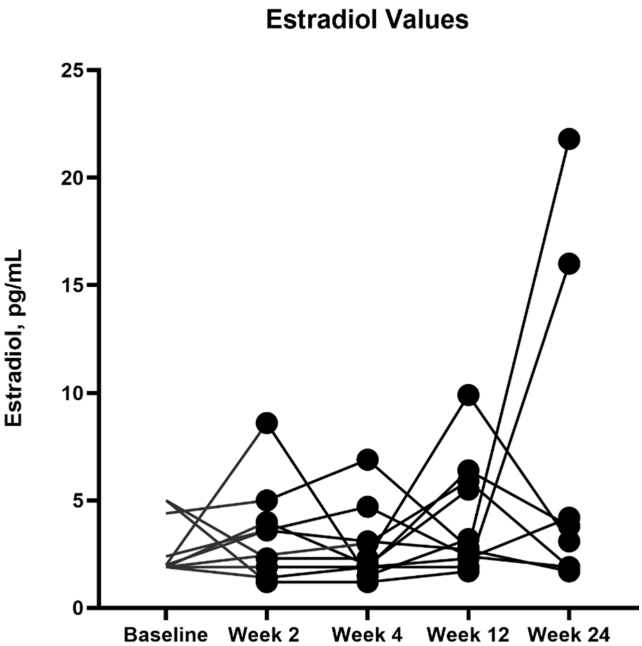

Fig. 2 Estradiol values during vaginal estrogen treatment

Importantly, serial measurement of serum estradiol using an ultra-sensitive LC-MS/MS assay demonstrated that estradiol levels remained within prespecified safety thresholds during the first 12 weeks of treatment in all participants and remained low in the majority of patients throughout the study period. These findings provide prospective evidence that low-dose vaginal estrogen can meaningfully improve genitourinary symptoms while producing minimal systemic estrogen exposure in most patients receiving aromatase inhibitor therapy. Given longstanding concerns that local estrogen therapy may counteract aromatase inhibitor–mediated estrogen suppression, our results contribute biologic and clinical data to an area where randomized prospective evidence has been limited. This trial also included 26% women who were premenopausal on ovarian suppression and 57% patients had stage II hormone positive cancers, providing evidence in these understudied patient subsets. An often-cited 2006 trial [20] involving seven women using Vagifem® 25 mcg vaginal tablets showed a more significant spike in estradiol level, with levels going from ≤ 5 pmol/L at baseline to a mean of 72 pmol/L at 2 weeks. This spike was significantly higher than what we observed on our trial. The Kendall trial also found that two women had persistently elevated estradiol level in weeks 7–10, at 219 and 137 pmol/L. The differences between this trial and the present one may reflect both lower estradiol dosing in the present study (10 mcg vs. 25 mcg) and the improved analytic specificity of LC-MS/MS compared with earlier extraction-based immunoassays. In most current studies, liquid chromatography-tandem mass spectrometry (LC-MS/MS) estradiol quantitation is widely considered the gold standard, especially for measuring very low levels of estrogen, where RIA has been found to often give falsely high values [21, 22].

Another study also showed a significant increase in estradiol level for patients using vaginal estrogen (either a vaginal ring or 25 mcg vaginal tablet) while on adjuvant tamoxifen or aromatase inhibitor therapy [23]. For patients using the vaginal ring, estradiol levels 12 weeks after ring insertion were 30 pmol/L higher than baseline levels, on average. Similarly, at 12 h after vaginal tablet insertion, subjects demonstrated estradiol levels 76 pmol/L higher than baseline. Again, this study also utilized the RIA assay for estradiol measurement, and they used a higher dose of the vaginal tablet.

We have accumulated more data in recent years to indicate that vaginal estrogen may be reasonable and very low risk for most breast cancer patients. A recent meta-analysis found that vaginal estrogen in breast cancer survivors was not associated with an increased risk of breast cancer recurrence or mortality [24].

A phase II randomized trial of 0.005% estriol vaginal gel versus placebo for women on AI therapy showed that vaginal estrogen was effective in terms of patient reported symptoms, improvement in vaginal pH, and vaginal maturation scoring. Serum estriol levels saw a transient increase at week 1 from a median of 0.5 pg/mL to 3.9 pg/mL at 1 week, with quick resolution to baseline levels thereafter [25].

Previous trials have also examined the efficacy of vaginal estrogen in this population of women taking adjuvant aromatase inhibitor therapy for breast cancer. Streff, et al. studied the vaginal ring in 8 patients prospectively and 6 patients retrospectively. The study found the estrogen-releasing vaginal ring to be safe and effective, with no significant increase between baseline and 16-week serum estradiol levels [26]. Another study examined twelve weeks of 10 mcg vaginal suppository (Vagifem®) in 20 patients on adjuvant letrozole therapy who were experiencing GSM. They did find 17 of 20 patients had either transient or persistent estradiol elevations, with mean of 8.8 pmol/L at 2 weeks, and 4.6 pmol/L at 4 weeks. The highest estradiol value was 110 pMol/L in one patient at 8 weeks. At all other time points before and after, this patient was suppressed with estradiol level < 5 pmol/L [27].

The clinical significance of these relatively mild, transient increases in estrogen level remain unclear. Especially when taken in the context of other common clinical scenarios which may cause unintended, more persistent, elevations of estradiol levels, we may have to accept that estradiol is frequently not ideally suppressed in certain populations of patients. For instance, obese post menopausal women on aromatase inhibitor therapy tend to have higher estrogen levels than their non-obese counterparts [28, 29].

Furthermore, particularly in these overweight and obese patients, the choice of AI may also alter estradiol suppression [28]. Pre-menopausal women who are taking GnRH agonist therapy with an AI, particularly those who have not received chemotherapy, are also at risk for incomplete ovarian suppression, commonly known as ovarian escape. In the SOFT-EST study, they report that at least 17% of patients on the SOFT trial experienced ovarian escape at some point during their first year of ovarian suppression therapy [30]. Additionally, and perhaps most relevantly, poor compliance with AI therapy is unfortunately quite common, with up to half of women non-adherent with five years of AI therapy. Clearly, AI non-adherence, especially if persistent, would increase the estradiol level; furthermore, this non-adherence is associated with increased mortality [31, 32]. In fact, outcomes may be ultimately improved if vaginal estrogen and other QOL interventions help improve compliance with AI therapy.

We must also consider these results in light of recent randomized and observational data which suggest that estrogen-only hormone therapy does not increase – and may in some setting reduce - breast cancer incidence [33–35]. With this understanding, small, transient rises in serum estrogen levels are less likely to be concerning in terms of recurrence risk.

While other trials have examined vaginal estrogen in breast cancer survivors, this study is unique in that it examines objective measures, such as vaginal pH and ultrasensitive serum estradiol levels, as well as more subjective patient-reported outcomes, such as the perception of dryness, dyspareunia, libido, and orgasm.

This study has several limitations. First, the small sample size and short follow up time do limit large generalizations about the safety of vaginal estrogen therapy. Additionally, related to the small sample size, only a subset of seven patients in the vaginal estrogen arm were sexually active and therefore able to answer the questions about sexual health. Among these seven patients, only six chose to answer the questions about libido and orgasm. This does limit the significance of our findings, given the small number of patients with analyzable data for these questions. Secondly, aromatase inhibitor compliance was not accurately measured on the trial, though we know that the two patients who had rise in estradiol level at the end of the trial had become noncompliant with AI therapy. If medication compliance had been measured consistently on the trial, we would be able to more clearly correlate AI adherence with estradiol levels. Finally, the open-label design may have influenced patient-reported outcomes.

In summary, this randomized study demonstrates that low-dose vaginal estrogen provides superior relief of genitourinary symptoms compared with a non-hormonal moisturizer in breast cancer survivors receiving aromatase inhibitor therapy, while systemic estradiol levels remained largely suppressed when measured using a highly sensitive LC-MS/MS assay. Although transient or late estradiol elevations were observed in a small number of patients, these findings occurred in the setting of reported non-adherence to endocrine therapy and highlight the multifactorial determinants of systemic estrogen exposure in this population. The clinical significance of modest biochemical estradiol fluctuations remains uncertain; however, improving quality of life and sexual health may play an important role in supporting long-term adherence to aromatase inhibitor therapy. Given the small sample size and limited duration of follow-up, larger prospective studies are needed to better define the long-term endocrine and oncologic safety of vaginal estrogen therapy. Nevertheless, these data provide prospective biologic reassurance that carefully selected patients receiving aromatase inhibitors may achieve meaningful symptomatic benefit from local estrogen therapy with minimal systemic exposure.

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Author contributions Study conception and design was performed by KO, PN, MR, and JC. Material preparation and data collection were performed by AB, CF, HA, MS, AA, AM, and EAN. Statistical design and analysis was performed by SM. Study procedures were performed by PN, KS, HM, AP, JC, KO, MR, JN, AV, DA, RH, and TZ. The first draft of the manuscript was written by PN, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets supporting the conclusions of this study were stored in a secured shared drive and will be shared by the corresponding author upon reasonable request. The data generated in this study are not publicly available due to restrictions related to patient privacy, informed consent, and institutional review board (IRB) requirements. Requests for access to de-identified individual participant data may be considered on a case-by-case basis, subject to institutional approvals and data use agreements.

Declarations

Ethical approval This study was conducted in accordance with principles of Good Clinical Practice. The study protocol was approved by the institutional review boards at Baylor College of Medicine and Houston Methodist Hospital (IRB number 00017710). The trial was registered at Clinicaltrials.gov (NCT 01984138). The trial was monitored by the Institutional Data and Safety Monitoring Committee at Houston Methodist Hospital to ensure participant safety and adherence to ethical and regulatory standards.

Consent to participate Written informed consent was obtained from all patients before participation in the study.

Competing interests The authors declare no competing interests.

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